Master Data Structures

with

Loop & Logic

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Data Structures Syllabus First 3 Classes		
Class No.	**Topic**	**Details**
1st Class	Introduction to Data Structure	**[**👉 এই অংশের জন্য কোনো **prior coding knowledge** লাগবে না**]** • Course outline discussion • Why we need data structures • Real-life examples • How computer memory stores data • Dynamic memory allocation • Static memory allocation • Difference between dynamic & static memory allocation
2nd Class	Prerequisite Topics from C Programming	**[**👉 **C Programming** এর এই **basics** গুলো ছাড়া আপনাকে অনেক **struggle** করতে হবে**]** • Array • Pointer • Function • Structure • Self-referential Structure
3rd Class	Stepping into the Data Structure Paradigm	**[**👉 এই ক্লাসে আমরা **Data Structure** এর মূল ধারণা ও **Linked List** এ **focus** করবো**]** • Linked List • Node creation • Insertion • Deletion

4th Class	Linked List - 1 (without function)	• **Creating Linked List**
		• **Insert at**
		  ○ **End**
		  ○ **Beginning**
		  ○ **Any position**
		• **Searching through Nodes**

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5th Class	Linked List - 2
	(without function)
		• **Creating Linked List**
		• **Delete at**
		  ○ **End**
		  ○ **Beginning**
		  ○ **Any position**

6th Class	Linked List - 3
	(with function)
		• **Creating Linked List**
		• **Insert at**
		  ○ **End**
		  ○ **Beginning**
		  ○ **Any position**
		• **Searching through Nodes**

7th Class	Linked List - 4
	(with function)
		• **Creating Linked List**
		• **Delete at**
		  ○ **End**
		  ○ **Beginning**
		  ○ **Any position**

8th class	Doubly Linked List	**Everything related to Double Linked List** **[It will not take much time if you have covered Singly Linked Lists well]**
9th Class	**Question Solve**	
10th Class	Stack (Implementation through Array)	• **Stack creation** • **Push()** • **Pop()** • **Peek()** • **Search()**
11th Class	Stack (Implementation through Linked List)	• **Stack creation** • **Push()** • **Pop()** • **Peek()** • **Search()**
12th Class	Expression Evaluation using Stack	• **Infix** • **Prefix** • **Postfix**
13th Class	Queue (Implementation through Array)	• **Queue creation** • ***Enqueue()*** • **Dequeue()** • **Peek()** • **Rear()** • **Search**
14th Class	Stack (Implementation through Linked List)	• **Queue creation** • ***Enqueue()*** • **Dequeue()** • **Peek()** • **Rear()**

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		• **Search**
15th Class	**Question Solve**	
16th Class	Introduction to Non Linear Data Structure & Basics of Tree	• **Introduction** • **Keywords** • **Type of Trees**

17th Class	Binary Tree	• **Implementation**
		• **Representation**
		• **Tree Traversal**
		  ○ **Inorder**
		  ○ **preorder**
		  ○ **postorder**
		• **Construct tree from traversal order**

18th Class	Binary Search Tree	• **BST** • **BST insertion , deletion** • **BST code**
19th Class	AVL	• **AVL** • **AVL insertion, deletion** • **AVL code**
20th Class	**Question Solve**	
21th Class	Heap Tree	• **Heap sort** • **Heap Code**

22th Class	Graph	• **Graph Representation**
		• **Graph Traversal**
		  ○ **BFS**
		  ○ **DFS**

23rd Class	Graph	**Extra**
24th Class	Question Solve	